DOMUS

Shared Memory Agent for Async Coordination

Challenge	Gemini Live Agent Challenge 2026
Category	Live Agents
Submission Deadline	March 16, 2026 at 5:00 PM CDT
Tech Stack	Gemini Live API · Google ADK · Firebase Firestore · Cloud Run
Team	Clarissa Chen + Developer Partner

PART 1 — PROJECT BRIEF

Background

Households and shared-responsibility partnerships run on shared context — doctor appointments, grocery lists, school schedules, medication reminders. When two partners operate on different schedules (working parent + stay-at-home parent, shift workers, co-parents), that context lives in people's heads and gets lost.

Existing tools — shared calendars, note apps, task managers like Asana or Google Keep — fail not because they lack features, but because they require intentional, friction-heavy data entry at the wrong moment. One partner is mid-errand, and can't stop to update an app. The other partner is between meetings and can't read a long message thread. The tools sit there waiting to be updated. Life does not wait.

Domus is a shared memory agent that lives between two partners. It captures context when you speak to it, sees physical documents when you hold them up, and proactively delivers the right information to the right person at the right moment. It doesn't wait to be updated — it asks, listens, and remembers.

Problem Statements

I am **a working parent**. I am trying to **stay informed and prepared for household events and decisions**. But I miss critical updates because my partner is too busy to text or update a shared app because we're both operating at full capacity with no overlapping bandwidth, which makes me feel ***anxious, unprepared, and out of the loop.***

I am **a stay-at-home parent / shift worker**. I am trying to **capture and share important household information in the moment**. But I have no easy way to log things like post-appointment notes or grocery needs because my hands are literally full and stopping to type isn't realistic, which makes me feel ***like I'm carrying the entire mental load alone.***

I am **both partners**. I am trying to **coordinate around recurring household rhythms (grocery run, appointments, school)**. But reminders are passive and easy to dismiss, and the right information never reaches the right person at the right time because every tool we've tried requires maintenance we don't have bandwidth for, which makes me feel ***like we're constantly playing catch-up with our own household.***

Goals

- Enable zero-friction, hands-free capture of household context via voice and camera
- Create a shared persistent memory that both partners can access and contribute to asynchronously
- Proactively surface the right information to the right partner at the right moment
- Learn household rhythms and patterns to become more anticipatory over time
- Demonstrate that AI agents can meaningfully reduce the mental load of household coordination

Non Goals

- Full household management suite — no chore tracking, budgeting, or finances in V1
- Recipe or meal planning engine (noted as future iteration)
- Calendar replacement or direct calendar integration
- Single-user personal assistant — this is explicitly a two-person shared system
- Mobile app — web app is sufficient for MVP and demo

Hypothesis

If we...

enable zero-friction voice + vision capture and proactive delivery of household context between partners...

Then...

coordination breakdowns ("I didn't know about that") will decrease and both partners will feel a measurable reduction in mental load...

Leading to...

higher household coordination satisfaction, fewer dropped tasks, and a demo experience that judges recognize as genuinely useful — not a tech demo for its own sake.

Vision Narrative

Tuesday morning. Sarah is at her desk, back-to-back meetings from 9am. Her husband Marcus is home with their 6-year-old, halfway through a math lesson when the pediatrician's office calls — Lily's annual checkup is Thursday at 2pm.

Marcus can't text Sarah right now. He's mid-lesson. He glances at his phone and says quietly:

🏠 **Marcus says (voice, hands-free):**

"Hey Domus — Lily's checkup is Thursday at 2pm. We've been meaning to ask about the vaccine schedule."

Domus logs it. Thursday is noted. The question is saved to the appointment thread. Marcus goes back to the math lesson.

Wednesday night, Sarah finally has a quiet moment. She asks:

💼 **Sarah says:**

"What's happening tomorrow?"

Domus briefs her in 30 seconds: appointment at 2pm, here are the two questions Marcus flagged, bring the insurance card.

Thursday. Marcus is in the waiting room. The doctor hands him a printed visit summary. He holds his phone up to the paper. Domus reads it, extracts the key information, marks the thread resolved.

Friday morning. Domus checks in proactively:

🛒 **Domus asks Marcus:**

"Grocery run is tomorrow. You mentioned you're low on almond milk on Monday. Anything else to add?"

Marcus answers out loud while making breakfast. List updated. Done. No app opened. No message sent. No mental load transferred.

Rough Scoping & Timeline

V1 — Hackathon MVP (4 days)

- Voice capture via Gemini Live API — hands-free, real-time, interruptible
- Camera capture of physical documents — prescriptions, appointment papers, pantry shelves
- Shared persistent memory — Firebase Firestore, two users joined to one household
- Proactive check-ins — agent initiates at rhythm-appropriate moments (not passive reminders)
- Briefing mode — "what do I need to know?" returns a structured summary
- Pantry/staples awareness — agent learns household defaults, flags what's missing
- Natural barge-in handling — agent pauses gracefully when interrupted, resumes context
- Household onboarding — set up recurring rhythms (grocery day, school schedule, etc.)

Future Iterations

- Recipe suggestions based on pantry staples + known picky eater preferences
- Grocery delivery integration — agent places the order directly
- Medication tracking and refill reminders
- School / homeschool schedule awareness
- Google Calendar integration
- Mobile app (iOS / Android)
- Multi-child and caregiver coordination (elder care use case)

Key Trade-Offs & Decisions

- **Category choice:** Live Agents category over Creative Storyteller or UI Navigator
  - The core interaction model is voice-first real-time, which aligns directly with the Live Agents definition. This is not a storytelling product or a screen-navigation product.

- **Shared state:** Firebase Firestore over custom backend for shared memory
  - Firestore provides real-time sync between two users out of the box, is a Google Cloud service (satisfies judging requirements), and dramatically reduces build time for a 4-day sprint.

- **Platform:** Web app over mobile for MVP

  - Faster to develop, easier to demo, and sufficient to prove the concept. Mobile is a natural next step but not required for the hackathon.

- **Demo scope:** Single demo thread (doctor appointment) over showing all features
  - Depth beats breadth in a 4-minute video. One complete, emotionally resonant story is more convincing than five half-demonstrated features.

PART 2 — PROJECT PROPOSAL

Architecture & Technical Proposal

Frontend	React web app (component-based, responsive)
Real-time Voice/Vision	Gemini Live API via WebSocket connection
Agent Orchestration	Google ADK (Python backend)
Shared Memory	Firebase Firestore (real-time sync, two-user household model)
Hosting	Google Cloud Run
Auth	Firebase Auth (household pairing model)

Division of Labor

- Clarissa owns: Frontend UI/UX (React), demo script, architecture diagram, video production, submission write-up
- Developer partner owns: Gemini Live API WebSocket integration, ADK Python backend, Firebase setup, Cloud Run deployment, proof-of-deployment recording

Core Agent Flows

1. Capture: Partner speaks + optionally shows camera. Gemini Live processes audio/video stream. ADK agent extracts structured data (event, note, grocery item) and writes to Firestore household document.
2. Query: Either partner asks "what do I need to know?" Agent reads Firestore, generates a prioritized briefing, responds via voice.
3. Proactive check-in: ADK agent monitors household rhythms (set during onboarding). At appropriate time, initiates conversation with relevant partner. Captures response. Updates Firestore.
4. Document capture: User holds physical document to camera. Gemini Live vision reads and extracts key information. Agent structures it and logs to the relevant thread.
5. Thread resolution: Agent marks a topic (e.g., appointment) as resolved after post-event capture. Removes from active briefing.

Risks & Mitigations

Risk	Likelihood	Mitigation
Gemini Live API learning curve exceeds estimated time	Medium	Developer partner owns this exclusively; Clarissa does not touch backend
Demo instability on submission day	Medium	Record a clean demo video early (Day 3); live demo is bonus, not primary
Shared auth between two users is complex	Low-Medium	Use Firebase Auth with a simple household pairing code (one partner creates, other joins)
Scope creep pulls focus from core flow	High	Strictly limit V1 to 6 core features; anything else goes on future iterations list
Proof of Cloud deployment missing from submission	Low	Developer partner records screen of Cloud Run console on Day 3 before final push

Open Questions

- What is the agent's distinct persona and voice? (Judges specifically evaluate this for Live Agent category)
- How do we handle household onboarding in the demo — do we show setup, or start with a pre-configured household?
- Do we show both partner views in the 4-minute demo, or focus on one partner's experience end-to-end?
- Should the briefing mode be voice-only or also display a visual summary on screen?
- How do we handle the bonus points? Writing a build blog post is low effort given Clarissa's background — worth doing.

Appendix: Research & Context

Why Existing Tools Fail

Tried by user: Asana, Google Keep, shared Notes app. Common failure mode: all require intentional data entry at the wrong moment. The tool is passive — it waits. Domus is active — it asks.

Target Users

- Working parent + stay-at-home parent
- Shift workers (opposite schedules, shared household)
- Co-parents (coordinating around children asynchronously)
- Any two people running a shared responsibility with minimal schedule overlap

Judging Criteria Alignment

Innovation & UX (40%)	Barge-in, proactive agent, voice + vision capture — genuinely beyond the text box
Technical (30%)	Gemini Live API + ADK + Firestore + Cloud Run — full Google Cloud native stack
Demo (30%)	Single emotional story arc, real software, architecture diagram, live proof
Bonus points	Blog post about build process (+0.6 pts) — low effort, high return

Future Iterations List

- Recipe suggestions based on pantry staples + picky eater preferences
- Grocery delivery integration
- Medication tracking and refill reminders
- School/homeschool schedule awareness
- Google Calendar integration
- Mobile app
- Elder care / caregiver coordination use case

Adapted from Asana's spec template. Built for the Gemini Live Agent Challenge 2026.